

Government College of Engineering, Jalgaon
(An Autonomous Institute of Government of Maharashtra)

Name :	PRN :
Class : L. Y. B.Tech Computer	Batch:
Subject : CO456U Cloud Computing Lab	Sem : VIII th
Course Teacher: Prof. S. D. Cheke	Academic Year : 2025-26
Date of Performance :	Date of Completion :

Practical No. 2

Aim: Deploy Nginx on aws/gcp instance and demonstrate how it can be exposed to outside world.

Theory :

Objective: To deploy a web server (Nginx) on a cloud virtual machine (AWS/GCP) and make it accessible over the internet using a public IP.

What is Cloud Computing?

Cloud Computing is the delivery of computing services like servers, storage, and networking over the internet.

What is Nginx?

Nginx is a high-performance web server used to serve websites and applications.

- Handles multiple users efficiently
- Used as web server, reverse proxy, and load balancer

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What is a Firewall / Security Group?

- Controls incoming and outgoing traffic
- Example: Allow HTTP (Port 80) for web access

Requirements

- AWS or GCP account
- Ubuntu instance (EC2 / Compute Engine)

- SSH client (MobaXterm / Terminal)
- Internet connection

Procedure / Steps

Part A: Launch Instance

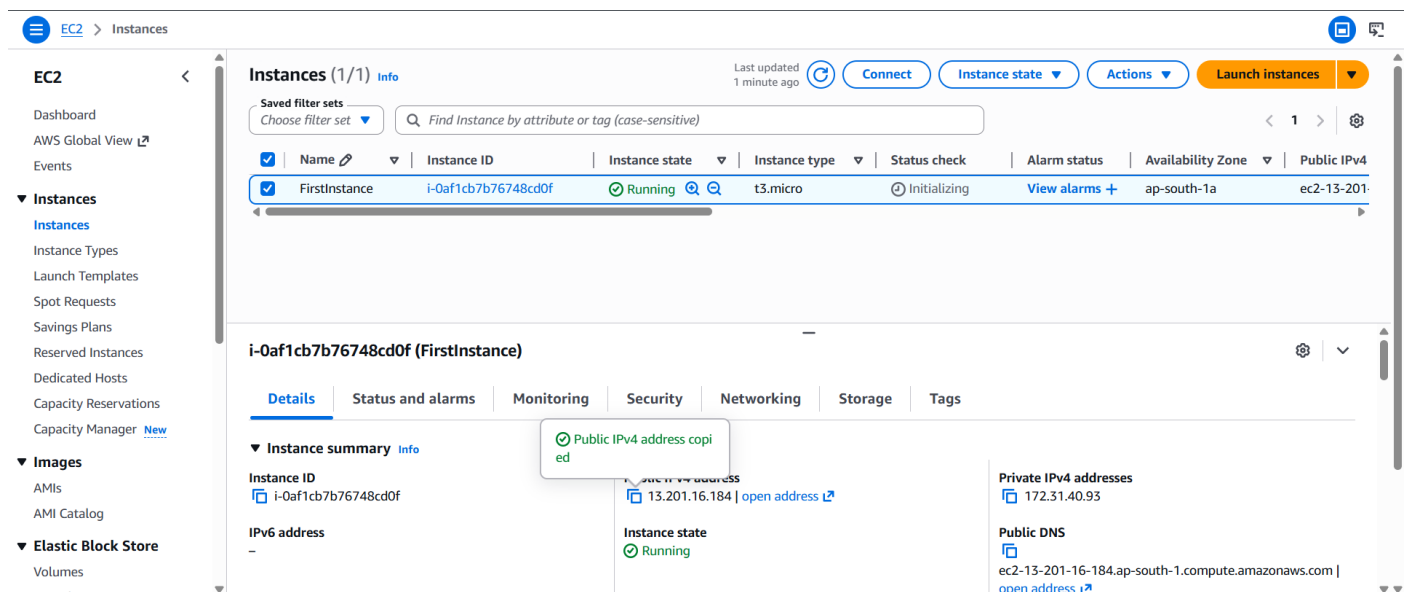
Step 1: Create Instance

1. Go to AWS EC2 / GCP Compute Engine
2. Click **Launch Instance / Create VM**
3. Select:
 1. OS: Ubuntu
 2. Instance Type: t2.micro (AWS) / e2-micro (GCP)
 3. Enable Public IP

Step 2: Configure Firewall

Allow:

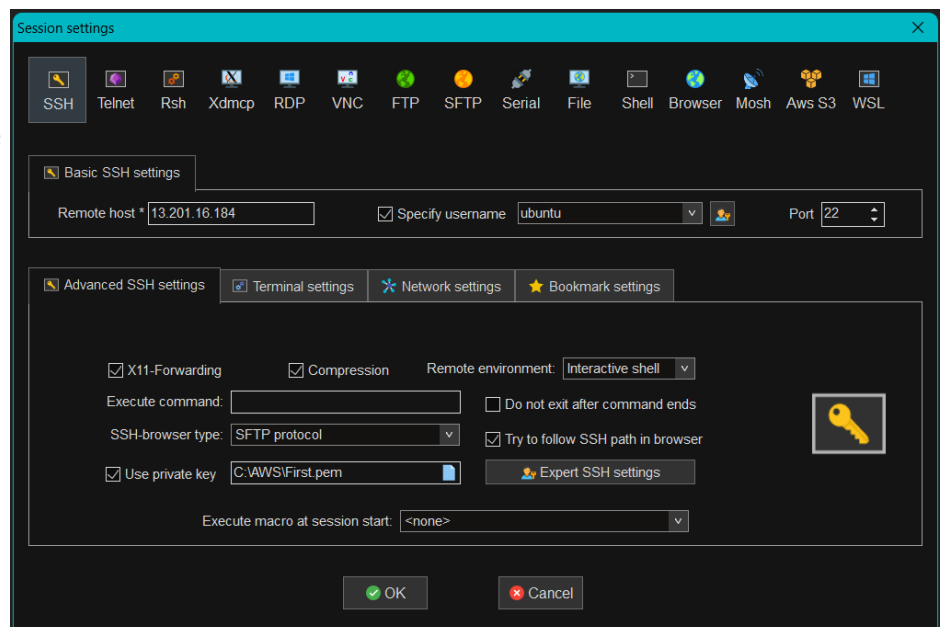
- SSH → Port 22
- HTTP → Port 80



Part B: Connect to Instance

Step 3: SSH Login

Use MobaXterm or terminal:



Part C: Install Nginx

Step 4: Update System

```
sudo apt update
```

Step 6: Start Nginx

```
sudo systemctl start nginx  
sudo systemctl enable nginx
```

Step 5: Install Nginx

```
sudo apt install nginx -y
```

Step 7: Check Status

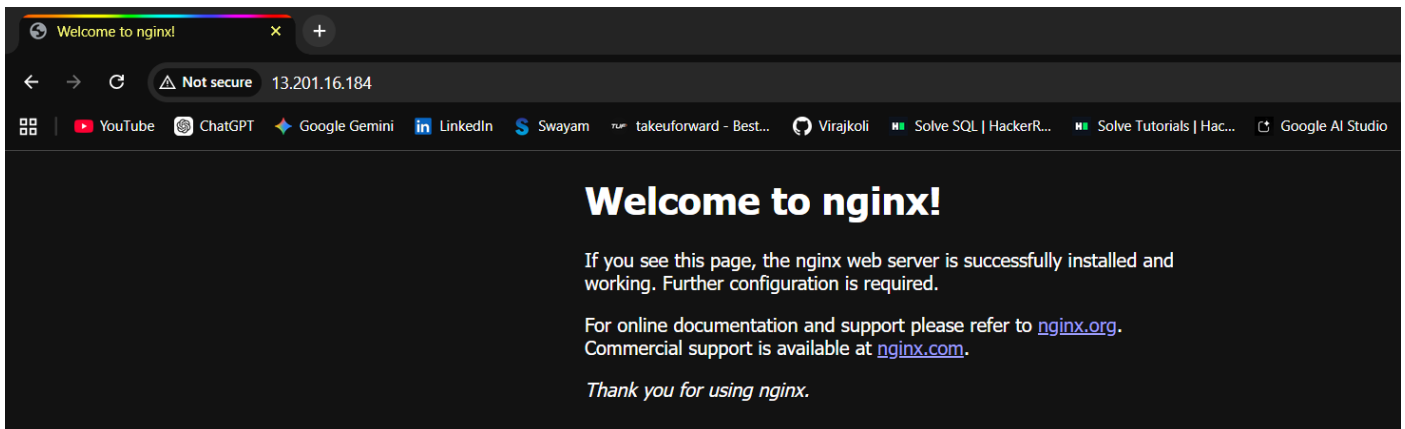
```
sudo systemctl status nginx
```

```
ubuntu@ip-172-31-40-93:~$ sudo systemctl start nginx  
ubuntu@ip-172-31-40-93:~$ sudo systemctl enable nginx  
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx  
ubuntu@ip-172-31-40-93:~$ sudo systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)  
   Active: active (running) since Thu 2026-04-16 17:33:39 UTC; 46s ago  
     Docs: man:nginx(8)  
  Main PID: 2703 (nginx)  
    Tasks: 3 (limit: 1004)  
   Memory: 2.4M (peak: 5.4M)  
      CPU: 25ms  
   CGroup: /system.slice/nginx.service  
           └─2703 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"  
             └─2705 "nginx: worker process"  
               └─2706 "nginx: worker process"  
  
Apr 16 17:33:39 ip-172-31-40-93 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...  
Apr 16 17:33:39 ip-172-31-40-93 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.  
ubuntu@ip-172-31-40-93:~$
```

Now, we the public ip address is exposed to browser through <http://your-public-ip>

Verify Output

- Default Nginx welcome page will appear
- This confirms server is publicly accessible



Conclusion: In this way, we have successfully implemented and completed the deployment of Nginx on a cloud instance and made it accessible to the outside world.

Prof. S. D. Cheke
Course Teacher